

Mikołaj Mieszko Charchuta

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Bioinformatician turning exploratory genomics into reproducible, production-grade pipelines (Nextflow, Snakemake, Docker, Conda), specialising in chromosome conformation capture (Meta Hi-C, 3C-seq) and machine learning on bacterial pangenomes. Bridging academia and industry: Meta Hi-C for phage-therapy R&D at Phagos, an independently grant-funded project on the 3D genome of *B. thuringiensis*, and prior R&D in personalised medicine and cellular agriculture. Three publications and invited teaching on reproducibility; EU remote-first, open to B2B contracts and industrial PhD (CIFRE) partnerships.

SKILLS

REPRODUCIBILITY
Nextflow • Snakemake • Docker
Conda • Git • CI/CD
Slurm • AWS

PROGRAMMING & ML
R • Python • Bash • C++
scikit-learn • tidymodels
R package development (NaïveR)

GENOMICS
3D genomics (Hi-C, Meta Hi-C, 3C-seq)
Pangenomics • WGS • WGBS
samtools • GATK • IGV • ChIP-seq

PHAGES & PROTEINS
Phage–host association (Meta Hi-C)
Toxin & gene mining in pangenomes
Protein structure (PyMOL)
Proteomics (PEAKS Studio, ProteinPilot)

TRANSCRIPTOMICS
DESeq2 • rMATS • rnaSPAdes

METAGENOMICS
QIIME2 • mothur • NaïveR

SEQUENCING & WET LAB
Nanopore • Illumina • Ion Torrent
Hi-C library prep • qPCR
DNA isolation • animal work

EDUCATION

BOKU, VIENNA
MSc Bioinformatics (Erasmus)
Sep 2026 -- Jul 2027
Academic exchange; thesis project with Prof. Christian Stauffer.

AMU, POZNAN
MSc in Bioinformatics
Oct 2025 -- Jul 2027
Thesis: *In silico analysis of Cry9 toxins and their genetic determinants in the Bacillus thuringiensis pangenome* (first-author article in preparation).

AMU, POZNAN
BSc Bioinformatics & Biotechnology
Oct 2021 -- Sep 2025
Thesis: *naïve methods for taxonomic classification from marker genes*, released as the NaïveR R package.

EXPERIENCE

PHAGOS | Bioinformatician
Jul 2026 -- Present | Remote / Paris, FR
- Deliver phage–host association calls into the therapeutic discovery pipeline by analysing Meta Hi-C data end-to-end, from raw reads to ranked pairings.
- Made analyses re-runnable and free of environment drift by rebuilding ad-hoc scripts as containerised Nextflow workflows (Docker, Conda) used across the R&D team.
- Guide two research projects and joint analyses with external academic partners (incl. Institut Pasteur), aligning experimental design with computational feasibility.

AMU, DEPT. OF MICROBIOLOGY | Contract Researcher
Sep 2025 -- Jan 2028 | Remote / Poznań, PL
- Won competitive Study@Research funding (app. nr. 201/34/UAM/0074) to run an independent project on the 3D genome of *B. thuringiensis*, owning Hi-C library prep, sequencing and analysis solo.
- Produced the dataset behind a first-author manuscript by mining the *B. thuringiensis* pangenome for Cry9 toxin determinants.
- Work on genomes of insect pests and biopesticidal microbes, combining comparative genomics with predictive protein design.
- Standardised the group's Hi-C/3C-seq analysis with reusable containerised workflows for contact-map QC and scaffolding.

GESELLSCHAFT FÜR BIOANALYTIK | Scientist
Jul 2024 -- Jun 2026 | Poznań, PL
- Delivered allergen and biosafety analyses to commercial turnaround and QA standards.
- Covered the whole chain from field sampling through lab work to computational analysis and client reporting.

BENE MEAT TECHNOLOGIES | Bioinformatician
Mar 2024 -- Jul 2024 | Prague, CZ
- Provided end-to-end bioinformatics support to cultivated-meat cell-line R&D.
- Ran transcriptomics (DESeq2, rMATS) and nanopore analyses, and automated bioreactor workflows.

LEIDEN UNIVERSITY MEDICAL CENTER | Bioinformatician
Jun 2023 -- Oct 2023 | Leiden, NL
- Contributed to two peer-reviewed publications by building the pipelines and figures for a murine MASLD study in the Endocrinology Department.
- Provided bioinformatics support to the lab and conducted animal research (mice, tissue and organ work).

MNM DIAGNOSTICS | Genome Analyst
Feb 2022 -- Jul 2022 | Poznań, PL
- Interpreted clinical whole-genome data for personalised oncology, reporting variants to clinicians.
- Supported the machine-learning team with feature extraction from methylation (WGBS) data, the data lake and explainable-AI (XAI) work.
- Automated tasks and data acquisition, and worked with biobanks under strict patient-data safety rules.

RESEARCH EXCHANGES

CHARLES UNIV., PRAGUE
Erasmus+
Winter 2022/23 • Summer 2023/24
• **Institute of Microbiology, CAS**: stable isotope probing of polar soil microbes for carbon tracking; meta-omics.
• **BIOCEV**: alternative splicing analysis (rMATS) on human RNA-seq data.
• CryoEM of cultivated-meat proteins.

UNIV. OF GDANSK & GUMED
National exchange
Winter 2023/24
• Metabolomics and proteomics of bioprotective synthetic communities (SynComs).
• Phage genomics.

GRANTS & AWARDS

• Study@Research, AMU (2026–27): 3D genome of *B. thuringiensis*
• SPUN international grant (2022–24): ML for soil biodiversity hotspots
• AMU grants for talented students: DynaMAP (2023–25), Trees of the Future (2021–23)
• Best poster, national metagenomics symposium

LANGUAGES

English (C2) • Polish (Native)
German (B2) • French (B2)
Czech (A2) • Hungarian (A2)

INTERESTS

Beekeeping & fungiculture • bikepacking • vegan cooking

RESEARCH PROJECTS

DYNAMAP NANOPORE PROFILING | Team lead, funded BSc thesis project
Oct 2023 -- Oct 2025 | AMU, Poznań
- Led a student team benchmarking a new nanopore-based microbiome profiling technique against established standards, showing its potential to lower sequencing costs.

SPUN GLOBAL SOIL MAPPING | International grant
Nov 2022 -- Apr 2024 | Illumina
- Predicted soil biodiversity hotspots with machine learning for a global mapping project (field video).

TREES OF THE FUTURE | Team grant
Oct 2021 -- Oct 2023 | Ion Torrent
- Identified interaction hubs between soil microorganisms; awarded best poster at the national metagenomics symposium; article in preparation.

NAÏVER | Open-source R package
github.com/mmcharchuta/NaiveR
- Naïve methods for taxonomic classification from marker genes, released from my BSc thesis.

TEACHING, TALKS & COMMUNITY

DANUBE INT. SCHOOL VIENNA | Lab Assistant
Jun 2026 -- Aug 2028 | Vienna, AT
- Design and run hands-on science labs; mentor students through independent research projects.

CRABS 2026 / HEALTH INTEL. | Invited Speaker & Workshop Lead
Aug 2026 | Kyiv, UA
- Lecture on *Reproducibility in Computational Genomics*; two Nextflow pipeline workshops; talk on Meta Hi-C & ML for bacterial genomes.

BIOINFORMATICS COMMUNITY PL | Co-organiser
2024 -- Present
- Event coordination, website and open-science advocacy.

PUBLICATIONS

In silico analysis of Cry9 toxins and their genetic determinants in the *Bacillus thuringiensis* pangenome. **Charchuta, M. M.** (first author), Baranek, J., et al. *In preparation, 2026.*

Combination of dietary fiber and exercise training improves fat loss in mice but does not ameliorate MASLD more than exercise alone. Kovynev, A.; **Charchuta, M. M.**; et al. *Am. J. Physiol. Gastrointest. Liver Physiol.*, 2025.

WED-517 Oatmeal a day does not keep the doctor away: inulin and exercise impact on MASLD amelioration and gut-liver crosstalk. Kovynev, A.; **Charchuta, M.**; et al. *J. Hepatol.*, 2024.

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